

Recap.

- ① $X \sim N(\mu, \sigma^2)$ is continuous distributed with a symmetric, bell-shaped density curve.
 $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, μ is the central of the curve, σ describes how wide the curve spreads.
 Use $\Phi(x)$ to express the cdf $F_X(x)$ when $X \sim N(0, 1)$.
- ② For a sequence of RVs $X_1, X_2, \dots, X_n$ independent from each other, let $S_n = \sum_{i=1}^n X_i$,
 with $E(S_n) = \sum_{i=1}^n E(X_i)$, $\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i)$, then $S_n \sim N(E(S_n), \text{Var}(S_n))$ and $Z_n = \frac{S_n - E(S_n)}{\sqrt{\text{Var}(S_n)}} \sim N(0, 1)$.
- ③ For a sequence of RVs $X_1, X_2, \dots, X_n$ and a RV X ,
 - (i) $X_n \xrightarrow{d} X$ if X_i has cdf of $F_i(x)$ and X has $F(x)$, $\lim_{n \rightarrow \infty} F_n(x) = F(x)$
 - (ii) $X_n \xrightarrow{P} X$ if $\lim_{n \rightarrow \infty} P(|X - X_n| \geq \epsilon) = 0$ for $\forall \epsilon > 0$.
 - (iii) $X_n \xrightarrow{a.s.} X$ if $P(\lim_{n \rightarrow \infty} X_n = X) = 1$.

Exercises.

Ex 1.

- (i) Independent: Passengers are individuals different from each other.
 Not-independent: Some passengers may related in social groups.
- (ii) Not computable without precise distribution, inequalities could give estimated result.
- (iii) $S_{100} = \sum_{i=1}^{100} X_i$, $E(S_{100}) = \sum_{i=1}^{100} E(X_i) = 100 \times 75 = 7500$, $\text{Var}(S_{100}) = \sum_{i=1}^{100} \text{Var}(X_i) = 100 \times 20^2 = 40000$.
 $Z_{100} = \frac{S_{100} - E(S_{100})}{\sqrt{\text{Var}(S_{100})}}$, $P(S_{100} > 8000) = P(Z_{100} > \frac{8000 - 7500}{\sqrt{40000}}) = 1 - P(Z_{100} \leq 2.5) = 1 - \Phi(2.5) \approx 0.6\%$.
- (iv) Profit per flight = # Passengers $\times$ Profit/Passengers = $100 \times 3 = 300 \text{ €}$.
 $E(\text{Cost of over capacity}) = P(\text{Over capacity}) \times \text{Cost} = 0.6\% \times 3000 = 18 \text{ €} < 300 \text{ €}$, profitable.

Ex 2.

- (i) $E(S_n) = \sum_{i=1}^n E(X_i) = 500 \times 10^4 = 5 \times 10^6$.
 $E(S_n) = nZ + \text{Expected profit}$, $Z = \frac{E(S_n) - \text{Expected profit}}{n} = \frac{5 \times 10^6 - 5 \times 10^5}{10^4} = 550$.
- (ii) With wlln, $\frac{S_n}{n} = \bar{X}_n \xrightarrow{P} \mu = 500$.
- (iii) To verify $P(S_n \leq nZ) \geq 0.95$, same as $P(S_n \geq nZ) \leq 0.05$.
 $\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i) = 6.4 \times 10^9$, $P(S_n \geq nZ) = P(S_n - E(S_n) \leq nZ - E(S_n))$
 $\leq P(|S_n - E(S_n)| \leq nZ - E(S_n))$
 $\leq \frac{\text{Var}(S_n)}{(nZ - E(S_n))^2} = \frac{6.4 \times 10^9}{(550 \times 10^4 - 5 \times 10^6)^2} = 0.0256$, sufficient.
- (iv) $P(S_n \geq nZ) = P\left(\frac{S_n - E(S_n)}{\sqrt{\text{Var}(S_n)}} \geq \frac{5.5 \times 10^6 - 5 \times 10^6}{\sqrt{6.4 \times 10^9}}\right) = 1 - \Phi(6.25) \approx 2 \times 10^{-10}$.

Ex 3.

(i) To prove: $\lim_{n \rightarrow \infty} P(|X_n + Y_n - (X + Y)| \geq \epsilon) = 0$.

With triangle inequality: $|(X_n - X) + (Y_n - Y)| \leq |X_n - X| + |Y_n - Y|$

(if left side $\geq \epsilon$, then at least one of the terms on the right side $\geq \frac{\epsilon}{2}$)

Define events $A: |X_n - X| \geq \frac{\epsilon}{2}$ $B: |Y_n - Y| \geq \frac{\epsilon}{2}$, then $A \cup B: |X_n + Y_n - (X + Y)| \geq \epsilon$, $P(A) = P(B) \rightarrow 0$ when $n \rightarrow \infty$.

$$\lim_{n \rightarrow \infty} P(|X_n + Y_n - (X + Y)| \geq \epsilon) = \lim_{n \rightarrow \infty} P(A \cup B) \leq \lim_{n \rightarrow \infty} (P(A) + P(B)) = \lim_{n \rightarrow \infty} (P(|X_n - X| \geq \frac{\epsilon}{2}) + P(|Y_n - Y| \geq \frac{\epsilon}{2})) = 0.$$

(ii) To prove: $\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = 0$.

With Markov's inequality: $\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \frac{E(|X_n - X|)}{\epsilon} \rightarrow 0$

(iii) Not discussed, see formal solution.

(iv) From (ii): with $E(|X_n - X|) \rightarrow 0$ and $E(|X_n - Y|) \rightarrow 0$, we imply $X_n \xrightarrow{P} X$ and $X_n \xrightarrow{P} Y$.

From (iii): with $X_n \xrightarrow{P} X$ and $X_n \xrightarrow{P} Y$, we imply $P(X = Y) = 1$.

Ex. 4.

(i) Characteristic function is fourier transform of density function, it is equivalent to prove that $\lim_{n \rightarrow \infty} \phi_{X_n}(t) = \lim_{n \rightarrow \infty} e^{i\mu_n t - \frac{1}{2}\epsilon_n^2 t^2} = e^{i\mu t - \frac{1}{2}\epsilon^2 t^2} = \phi_X(t)$.

(ii) Let $Y_n = X_n - \mu_n$, $Y_n \sim N(0, \epsilon_n^2)$.

With Chebyshev's inequality, when $n \rightarrow \infty$: $P(|Y_n| \geq \epsilon) = P(|X_n - \mu_n| \geq \epsilon) \leq \frac{\text{Var}(X_n)}{\epsilon^2} = \frac{\epsilon_n^2}{\epsilon^2} \rightarrow 0$.

Therefore $Y_n = X_n - \mu_n \xrightarrow{P} 0$, $X_n \xrightarrow{P} \mu_n$. Since $\mu_n \rightarrow \mu$, $X_n \xrightarrow{P} \mu$.